

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0258

Dewey 138 kV
October 17, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0258. The Interconnection Customer proposes a 30.6 MW solar photovoltaic facility interconnecting at the Dewey 138 kV substation in the Kentucky area. The request was received on August 08, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 185 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,810,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: KETTLE CRK PRIMARY - MT VERNON 138kV
Contingency: HUBBARDSTOWN - BAKER 345kV
Loading: 127.8% (Rating: 217 MVA)
DFAX: 33.5%
Pre-project loading: 123.0%
2. Facility: AMERICAN FALLS - REATA 138kV
Contingency: DOWELL - WEST PAINTSVILLE 138kV line
Loading: 120.6% (Rating: 221 MVA)
DFAX: 5.3%
Pre-project loading: 119.9%
3. Facility: GLENOVEER - REATA 138kV L
Contingency: DOWELL - WEST PAINTSVILLE 138kV
Loading: 119.6% (Rating: 224 MVA)
DFAX: 5.3%
Pre-project loading: 118.9%
4. Facility: KEAN MTN AND GRASSY CREEK 138/1kV Auto
Contingency: KEAN MTN AND GRASSY CRK - CLINCH RIVER 345kV
Loading: 118.6% (Rating: 345 MVA)
DFAX: 8.1%
Pre-project loading: 117.9%
5. Facility: GLENOVEER - GREENS BAYOU 69 138kV
Contingency: W PAINTSVILLE - DOWLL 138kV
Loading: 118.4% (Rating: 229 MVA)
DFAX: 5.3%
Pre-project loading: 117.7%

6. Facility: KETTLE CREEK PRIMARY - NEPTUNE SUBSTATION 138kV
Contingency: HBBARDSTOWN - BAKER SUBSTATION 345kV
Loading: 115.1% (Rating: 232 MVA)
DFAX: 33.5%
Pre-project loading: 110.7%
7. Facility: ASPEN 345-138 kV Transformer Bank
Contingency: CRANDALL - ASPEN 345kV line
Loading: 108.6% (Rating: 563 MVA)
DFAX: 5.2%
Pre-project loading: 108.3%
8. Facility: DILLARD-ORANGE - FISHKILL PLAINS 345kV
Contingency: DISMAL RIV - BECKER 345kV Line
Loading: 107.8% (Rating: 1,399 MVA)
DFAX: 14.0%
Pre-project loading: 107.5%
9. Facility: MEDORA SUBSTATION - HAMPTON RD 345kV
Contingency: FISHKILL PLAINS - SIMPSON STUART TAP 345kV Line
Loading: 104.7% (Rating: 1,243 MVA)
DFAX: 7.1%
Pre-project loading: 104.5%
10. Facility: MEDORA - P.T.S SUBSTATION 345kV
Contingency: RITTER PARK - SUGAR HILL 345kV
Loading: 104.4% (Rating: 1,258 MVA)
DFAX: 7.9%
Pre-project loading: 104.2%
11. Facility: RITTER PARK - SUGAR HILL 345kV
Contingency: HILLTOP - PINEY CREEK PROJECT 500kV
Loading: 103.5% (Rating: 1,250 MVA)
DFAX: 7.6%
Pre-project loading: 103.3%
12. Facility: KEAN MTN AND GRASSY CRK 345-1 kV Autotransformer
Contingency: KEAN MTN AND GRASSY CRK - CLINCH RIV 345kV
Loading: 103.4% (Rating: 396 MVA)
DFAX: 8.1%
Pre-project loading: 102.7%
13. Facility: SIDNEY COAL - MEEER 138kV line
Contingency: KETTLE CREEK PRIMARY - MT VERNON SUBSTATION 138kV
Loading: 102.0% (Rating: 197 MVA)
DFAX: 5.4%
Pre-project loading: 101.2%
14. Facility: ATOKA - SPRING CREEK RD 138kV
Contingency: CATAWBA - NEPTNE 345kV
Loading: 101.8% (Rating: 286 MVA)
DFAX: 5.2%
Pre-project loading: 101.2%
15. Facility: STEWART MTN DAM - RITTER PARK 345kV
Contingency: MEDORA - P.T.S 345kV
Loading: 101.8% (Rating: 1,173 MVA)
DFAX: 7.0%
Pre-project loading: 101.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KETTLE CRK PRIMARY - MT VERNON 138kV	Rebuild line with higher-capacity conductor	\$1,903,000	7% (\$127,000)
REATA - AMERICAN FALLS 138kV	Rebuild line with higher-capacity conductor	\$3,959,000	2% (\$58,000)
GLENOVEER - REATA 138kV L	Rebuild line with higher-capacity conductor	\$8,686,000	2% (\$126,000)
KEAN MTN AND GRASSY CREEK 138/1kV Auto	Replace transformer with higher-rated unit	\$4,851,000	1% (\$54,000)
GREENS BAYOU 69 - GLENOVEER 138kV Line	Rebuild line with higher-capacity conductor	\$2,554,000	2% (\$37,000)
KETTLE CREEK PRIMARY - NEPTUNE SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$33,666,000	7% (\$2,252,000)
ASPEN 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$13,433,000	1% (\$147,000)
DILLARD-ORANGE - FISHKILL PLAINS 345kV	Reconductor line	\$44,770,000	1% (\$361,000)
MEDORA SUBSTATION - HAMPTON RD 345kV	Reconductor line	\$9,407,000	1% (\$97,000)
MEDORA - P.T.S SUBSTATION 345kV	Reconductor line	\$12,599,000	1% (\$105,000)
RITTER PARK - SUGAR HILL 345kV	Reconductor line	\$9,740,000	1% (\$98,000)
KEAN MTN AND GRASSY CRK 345/1kV XFMR	Replace transformer with higher-rated unit	\$13,892,000	1% (\$156,000)
MEEER - SIDNEY COAL 138kV Line	Reconductor line	\$2,285,000	1% (\$29,000)
ATOKA - SPRING CREEK RD 138kV	Reconductor line	\$7,785,000	1% (\$94,000)
STEWART MTN DAM - RITTER PARK 345kV	Reconductor line	\$6,645,000	1% (\$69,000)
	Total allocated to project		\$3,810,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.